1. Coordination and Response

1.1. Mammalian Nervous System

- The mammalian nervous system consists of two parts:
  - **Central Nervous System** (CNS) consists of the brain and spinal cord, which are the areas of coordination.
  - **The peripheral nervous system** (PNS) is comprised of nerves and neurones that coordinate and regulate body functions.
- Electrical impulses travel through the neurones.
- The nervous system helps with the coordination and regulation of body functions.

1.2. Types of Neurones

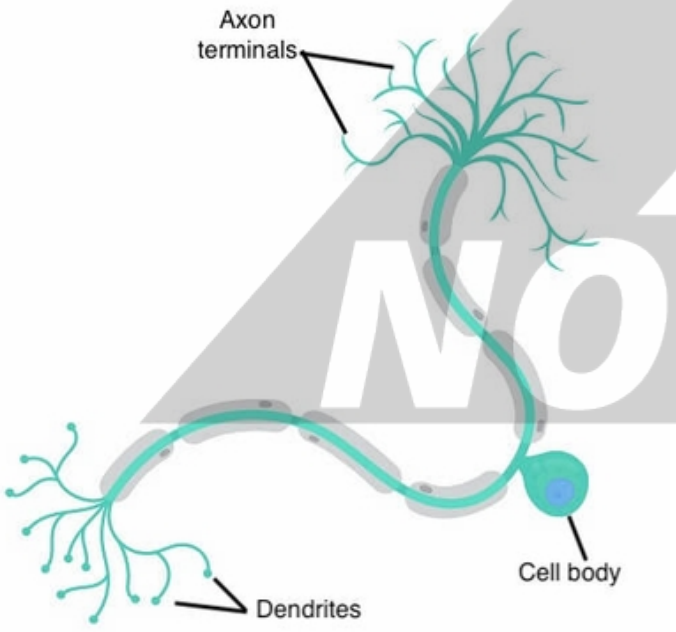
Nerve Impulse: an electrical signal that passes along the nerve cells called **neurones**

Axons: Ensure less time is wasted when the impulse is transferred from one neurone to another

Dendrites: Extensions which form a network for easy communication.

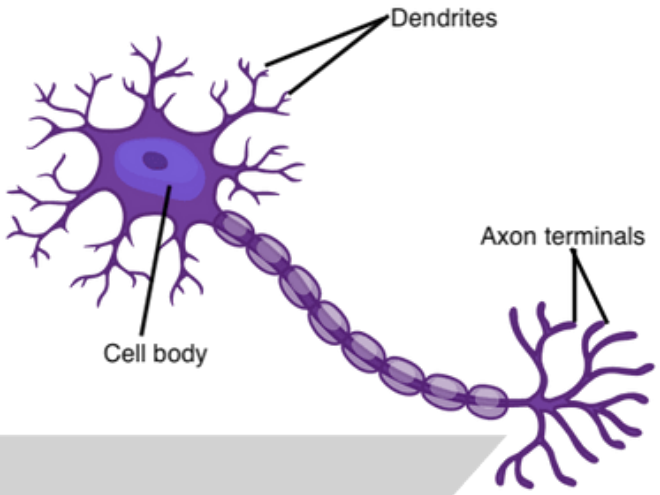
Sensory Neurone

- Carry impulses from the **sense organs (sensory receptors) to the CNS**.



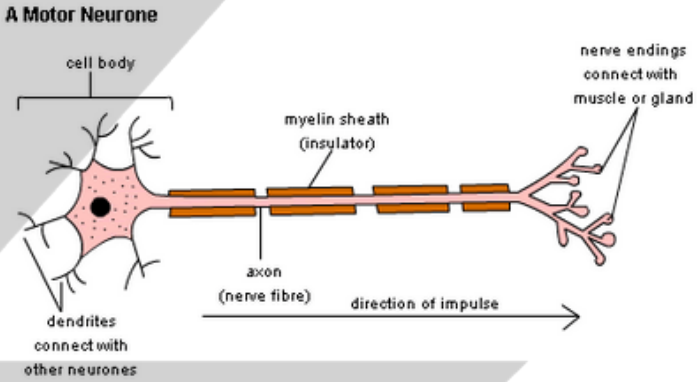
Relay Neurone (Interneurone)

- The middleman inside the CNS connects sensory to motor neurones.



Motor Neurone

- Carry impulses from the **CNS to effectors** (muscles or the glands)



1.3. Simple Reflex Arc

Reflex Action: automatically and rapidly integrates and coordinates the stimuli with the responses of effectors (muscles and glands).

- E.g. quickly removing your hand from the hot metal surface
- They involve three neurones: a sensory neurone, a relay neurone and a motor neurone.
- The gap between neurones is called a synapse.

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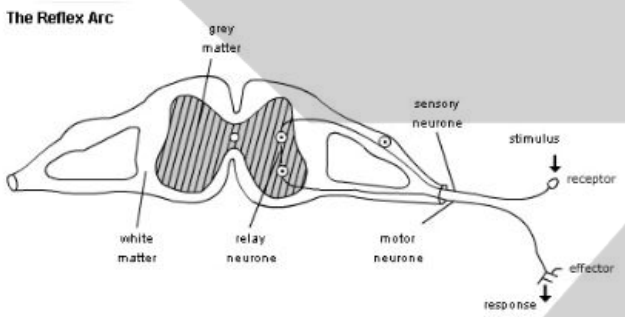
- How the simple reflex arc works:
  - A stimulus affects a receptor (cell or organ that converts a stimulus into an electrical impulse)
  - A sensory neurone carries impulses from the receptor to the CNS (spinal cord)
  - Connector/relay neurone carries impulse slowly (because it has no myelin sheath) across the spinal cord
  - The motor neurone carries impulses from the CNS to the effector
  - The effector (either a muscle or a gland) carries out the response

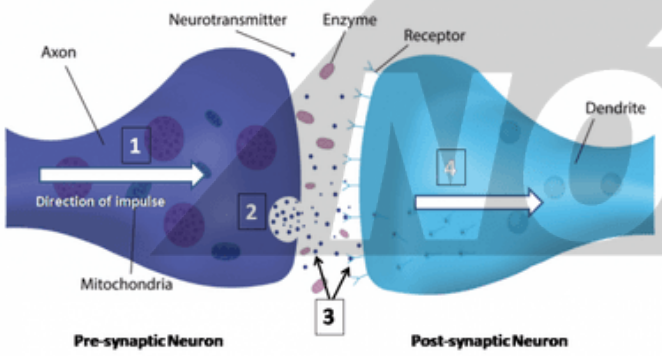
Flowchart of the reflex arc:

[Stimulus] → [Receptor (eg. in skin)] → [Sensory Neuron] → [Relay Neuron (in spinal cord)] → [Motor Neuron] → [Effector (e.g. Muscle)] → [Response (e.g. Pulling hand away)]



1.5. Synapse

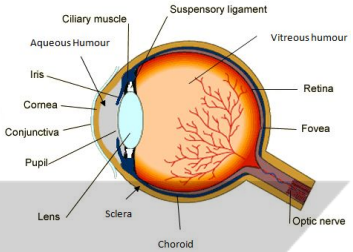
Synapse: a junction between two neurones, consisting of a gap across which impulses pass by diffusion of a neurotransmitter



- The synapses ensure that impulses travel in **one direction only**.
- **Synaptic cleft:** the small gap between each pair of neurones
- Inside the neurone's axon, there are 100s of tiny vacuoles (**vesicles**, each containing a chemical called neurotransmitter)
- When an impulse arrives, the vesicles move to the cell membrane and empty their content into the synaptic cleft.
- The neurotransmitter quickly diffuses across the tiny gap and attaches to receptor molecules in the cell membrane of the relay neurone.
- This can happen because the neurotransmitter molecules' shape complements the receptor shape.

1.7. Sense Organs

Sense Organ: groups of receptor cells responding to specific stimuli: light, sound, touch, temperature, and chemicals.



- **Cornea:** refracts light
- **Iris:** controls how much light enters the pupil
- **Lens:** focuses light onto the retina
- **Retina:** contains light receptors, some sensitive to light of different colours (Rods and cones)
- **Optic nerves:** carry impulses to the brain
- **Pupil**: A hole which allows light to enter

1.8. Pupil Reflex

- Adjusting for high and low light intensity
- An involuntary response

	Low Light Intensity	High Light Intensity
Radial Muscles	Contract	Relax
Circular Muscles	Relax	Contract
Pupil Size	Wide	Narrow
Amount of Light Entering	More	Less

1.9. Accommodation

Accommodation: Adjusting for near and distant objects.

- The way the **lens** brings about **fine focusing** is called **accommodation**
- The lens is **elastic,** and its shape can be changed when the **suspensory ligaments** attached to it become **tight** or **loose**
- The contraction or relaxation of the ciliary muscles brings about the changes

	Near Object	Distant Object
Ciliary muscles	Contract	Relax
Suspensory Ligaments	Slack	Tighten
Lens	Short and fat	Long and thin

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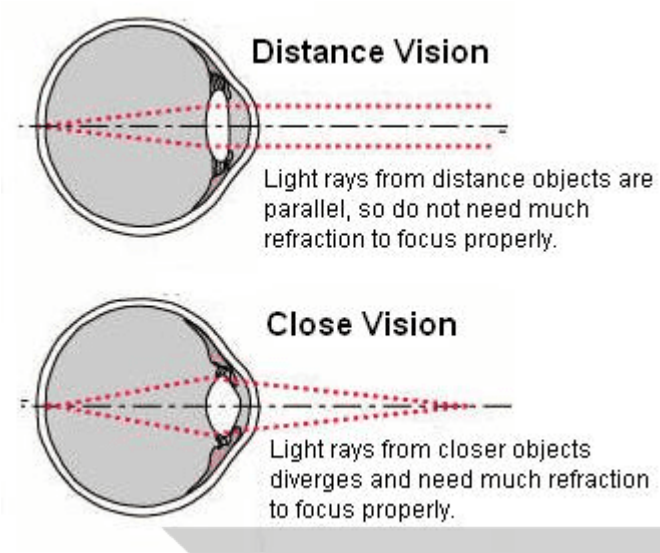


Example Past Year Question

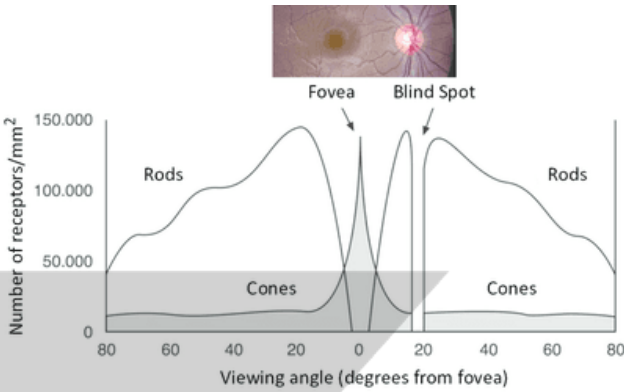
Explain why a person cannot focus on distant objects if the suspensory ligaments become permanently overstretched. (0610/42/F/M/23)

1. ciliary muscles relax
2. suspensory ligaments can no longer become tight
3. the lens is not stretched/remains wide
4. the angle of refraction remains unchanged

1.10. Rods and Cones

Rods	Cones
Provide low detail, black & white images, suitable for seeing in low-intensity light (at night).	Provide detailed, coloured images; they work in high light intensity.
Packed most tightly around the edge of the retina, so you can see things most clearly when not looking directly at them.	Most tightly packed at the retina's centre, objects are seen most clearly when directly looking at them.

- Fovea:
  - Part of the retina where the receptor cells are pushed most closely together
  - Where light is focused when you look straight at an object
- Distribution of Rods and Cones
  - More rod cells than cone cells
  - A number of cone cells peaked in the middle, showing the location of the fovea. No rod cells are present in the peak.
  - No rod and cone cells at the blind spot, as the optic nerve is located there. \n



1.11. Hormones

Hormones: A chemical substance produced by a gland and carried by the blood, altering the activity of one or more specific target organs.

Endocrine Glands

- adrenal glands and adrenaline
- pancreas and insulin
- testes and testosterone
- ovaries and oestrogen

Adrenaline

- A hormone secreted by the adrenal gland.
- It increases pulse rate, heart rate and pupil diameter.
- Increases blood glucose concentration for respiration.
- Adrenaline is secreted, for example, bungee jumping or riding a rollercoaster.

Gland	Hormone	Function
Adrenal gland	Adrenaline	Prepares the body for vigorous action
Pancreas	Insulin	Reduces the concentration of glucose in the blood
Testes	Testosterone	Causes the development of male sexual characteristics
Ovary	Oestrogen	Causes the development of female sexual characteristics
Pancreas	Glucagon	Increases concentration of glucose in the blood

1.12. Homeostasis

Homeostasis: The maintenance of a constant internal environment.

- Insulin decreases blood glucose concentration.

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- The concept of homeostatic control by negative feedback with reference to a set point

Negative Feedback Concept

Negative Feedback: controls the production of hormones and regulates their own production

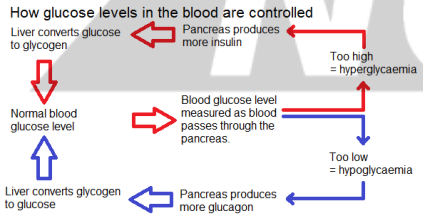
- A negative feedback control is when the change in hormone level acts as a signal to cancel out that change, so when the blood hormone level is low, hormone production is stimulated; when it is high, it is inhibited.

1.13. Nervous and Hormonal Control

Comparison	Nervous system	Endocrine system
Speed of action	Very rapid	Can be slow
Nature of message	Electrical impulses travelling along nerves	Chemical messengers (hormones) travelling in the bloodstream
Duration of response	Usually within seconds	It may take years (puberty)
Area of response	Localized response (only one area usually)	Widespread response (in many organs)
Example of process-controlled	Reflexes such as blinking	Development of the reproductive system

1.14. Homeostasis: Blood Glucose Control

- Blood glucose levels are monitored and controlled by the pancreas
- The pancreas produces and releases different hormones depending on the blood glucose level
- Insulin is released when blood glucose levels are high – the liver stores excess glucose as glycogen
- Glucagon is released when blood glucose levels are low – the liver converts stored glycogen into glucose and releases it into the blood



- When the control of blood glucose does not work, a person is said to have diabetes

Type 1 Diabetes

Type 1 Diabetes: caused by the death of the cells that secrete insulin.

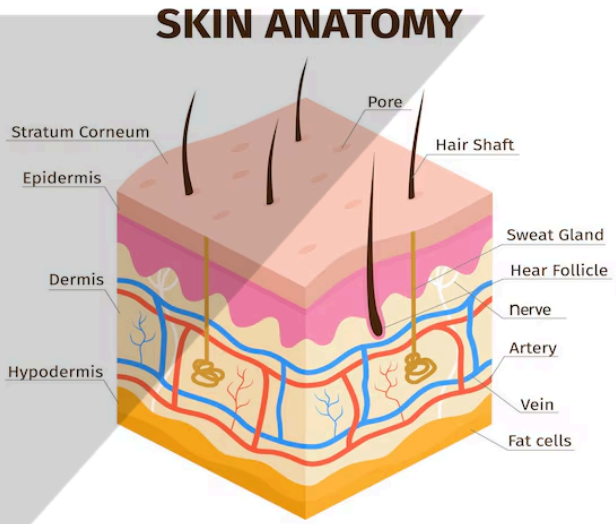
- Symptoms: hyperglycaemia (feeling unwell, dry mouth, blurred vision, and thirsty) or hypoglycaemia (tired, showing confusion and irrational behaviour)
- Treatment: eating little and often and avoiding large amounts of carbohydrates, injecting insulin to reduce blood glucose concentration

1.15. Homeostasis: Temperature Regulation

Control of body temperature is a homeostatic mechanism

- The human body maintains the temperature at which enzymes work best, around 37°C
- If body temperature increases over this temperature, enzymes will denature and become less effective at catalysing reactions such as respiration

The Structure of the Skin



Temperature Regulation by The Skin

- Regulation is controlled by the brain which contains thermoreceptors (sensitive to the temperature of the blood)
- The skin also has thermoreceptors and sends nervous impulses to the brain via sensory neurones
- The brain responds to this information by sending nerve impulses to the motor neurones, then to the effectors in the skin to maintain the temperature within a narrow range of the optimum, 37°C
- Fatty tissue under the dermis acts as a layer of insulation to prevent too much body heat from being lost through the skin

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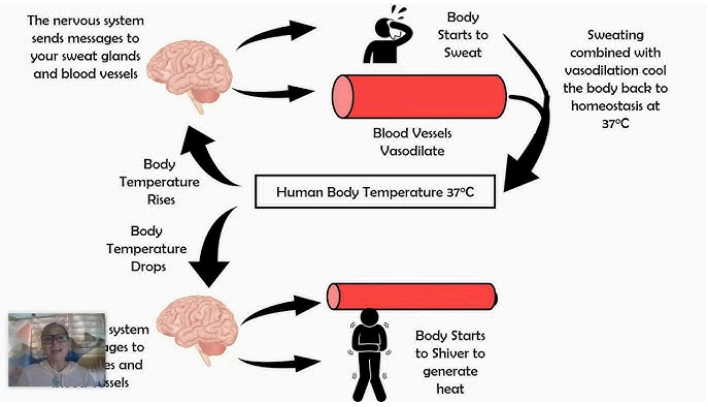


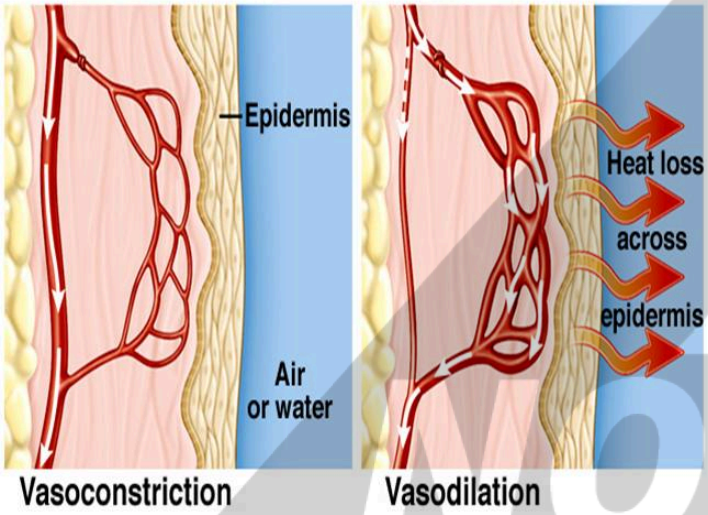
Vasoconstriction and Vasodilation

Vasoconstriction: blood flow in capillaries slows down because arterioles leading to the skin capillaries get narrower when we are **cold**.

- This reduces the amount of heat lost from the blood by radiation as less blood flows through the surface of the skin

Vasodilation: blood flow in capillaries increases because blood vessels to the skin capillaries get wider when we are **hot**.

- This cools the body as blood (which carries heat around the body) flows at a faster rate through the skin's surface and so more heat is lost by radiation



1.16. Tropic Responses

- **Auxin:**
  - Plant hormones or growth substances.
  - Controls tropisms
  - It is produced by cells at the tip of the roots and shoots of plants

Gravitropism: a response in which a plant grows towards (positive) or away (negative) from gravity.

- **Auxins' role in gravitropism:**
  - Made in the shoot tip
  - Then, it diffuses through the plant from the shoot tip
  - Auxin is unequally distributed in response to light and gravity
  - Auxin stimulates cell elongation

Phototropism: a response in which a plant grows towards (positive) or away (negative) from the direction light is coming.

- **Auxins' role in phototropism:**
  - If the sun shines on the right side of a plant's shoot, auxins accumulate on the dark opposite left side.
  - Auxins accumulating makes cells on the left side grow faster than cells on the right.
  - When the left side of the shoot starts growing faster than the right side, the shoot will start to bend to the right side towards sunlight.

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